

Shido Nakajima

Plano, TX; 512-825-9581; Shido.Nakajima@UTDallas.edu; linkedin.com/in/shido-nakajima; shidonakajima.com

EDUCATION

University of Texas at Dallas

M.S., Biomedical Engineering

Jan 2025 – May 2027

GPA: 3.926 / 4.0

University of Texas at Dallas

B.S., Biomedical Engineering

Aug 2020 – May 2024

GPA: 3.772 / 4.0

TECHNICAL SKILLS

Tools: 3D CAD, FEBio, Meshlab, 3D Slicer, Gantt Chart

Machine Learning: PyTorch, TensorFlow, MATLAB Deep Learning Toolbox

Computer Language: Python, MATLAB, Bash, Powershell, HTML, CSS, Javascript

PROFESSIONAL EXPERIENCE

Texas Biomedical Device Center, UT Dallas

Mar 2025 – present

Software Developer, Research Assistant

- Developed a peripheral nerve stimulation device, enabling synchronized hardware communication, data analysis, and stimulation
- Analyzed electromyography data of vocalization muscle, isolating motor unit signals along 64 channels

Neuronal Networks and Interfaces Laboratory, UT Dallas

Jan 2021 – Mar 2023

Research Assistant

- Analyzed neural activity in rat motor cortex through implanted microelectrode arrays, improving signal stability duration by 8-fold
- Implemented Butterworth filter on neuronal data, reducing noise by 30%
- Handled, habituated, anesthetized, and injected rats to observe change in collection of neuronal activity

PROJECTS

Sleep Structure Study Using Oura Ring

- Analyzed continuous sleep structure data over 2 months period, resulting in machine learning models with R-squared of 0.74
- Implemented an ETL pipeline on sleep structure data, reducing misdiagnosis of sleep stage by 70%
- Deployed a predictive model to generate sleep debt data, increasing data volume by 8-fold

Deep Learning Model for Age Prediction with Brain MRI

Collaborative work with UT Southwestern

- Developed a deep learning model that predicts age from brain MRI volume, resulting in accuracy of +/- 5 years
- Implemented an end-to-end automated pipeline for data preparation, visual feature extraction, and regressor age prediction, enabling easy deployment and reproducibility

Bleeding Control Trainer with Augmented Reality Interface

Capstone project

- Developed a Unity/C# based AR application for traumatic hemorrhage treatment training
- Managed the team using Gantt chart and Work Breakdown Structure, reducing development timeline by 2 weeks

PUBLICATIONS

Jeakle, E. N., et. al. (2023). Chronic stability of local field potentials using amorphous silicon carbide microelectrode arrays implanted in the rat motor cortex. *Micromachines*, 14(3), 680. <https://doi.org/10.3390/mi14030680>

ADDITIONAL INFORMATION

Eligibility: Eligible in the U.S. for internships & full-time – no restrictions

Language: English (native) and Japanese (native)